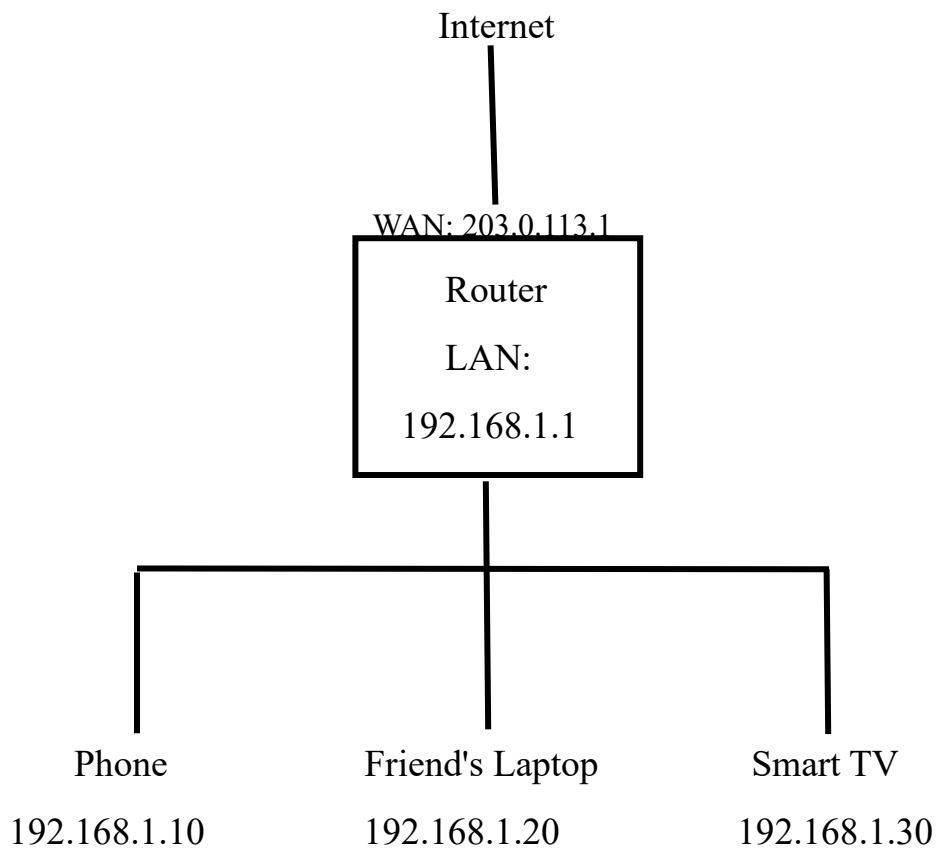


IP ROUTING AND ROUTING PROTOCOL

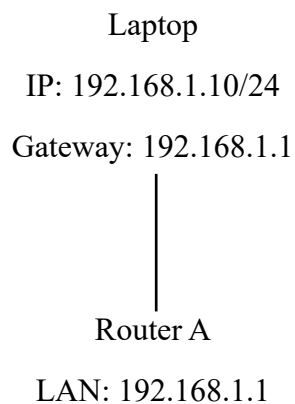
1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

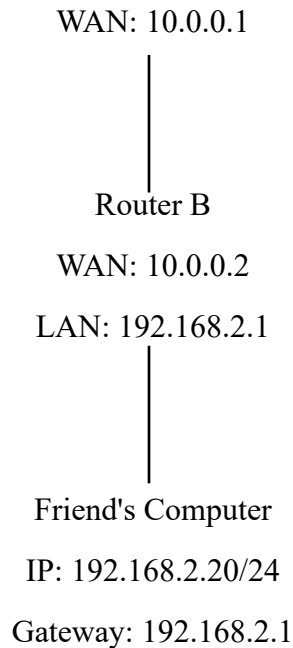


Device	IP Address
Router (WAN Interface)	203.0.113.1
Router (LAN Interface)	192.168.1.1
Phone	192.168.1.10
Friend's Laptop	192.168.1.20
Smart TV	192.168.1.30

**2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.**





Step 1

- The laptop (192.168.1.10) wants to send data to the friend's computer (192.168.2.20). Since the destination is in a different subnet, it sends the packet to its default gateway (192.168.1.1).
- **Router A** checks its routing table and forwards the packet to Router B (10.0.0.2).

Router A Routing Table

Destination	Next Hop
192.168.1.0/24	Direct
192.168.2.0/24	10.0.0.2

- **Router B** checks its routing table and finds that 192.168.2.0/24 is directly connected. It sends the packet to the friend's computer.

Router B Routing Table

Destination	Next Hop
192.168.2.0/24	Direct
192.168.1.0/24	10.0.0.1

- The friend's computer receives the packet and sends a reply back through Router B → Router A → Laptop.

Packet Path

Laptop (192.168.1.10)



Router A (192.168.1.1)



Router B (10.0.0.2)



Friend's PC (192.168.2.20)

This is how the packet travels between **two different subnets** using routers and routing tables.

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Scenario:

- Ahmedabad Branch → 192.168.1.0/24
- Surat Branch → 192.168.2.0/24
- Vadodara Branch → 192.168.3.0/24

Branch (City)	Network Address	Subnet Mask	Next Hop	Interface
Ahmedabad	192.168.1.0	255.255.255.0	Direct	LAN1
Surat	192.168.2.0	255.255.255.0	10.0.0.2	WAN1
Vadodara	192.168.3.0	255.255.255.0	10.0.0.3	WAN2

Explanation

- Ahmedabad Branch: Directly connected to the router through LAN1.
- Surat Branch: Packets are forwarded to the next hop 10.0.0.2 using WAN1.
- Vadodara Branch: Packets are forwarded to the next hop 10.0.0.3 using WAN2.

This routing table allows the three Zomato branches to communicate with each other over different subnets.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static Routing	Dynamic Routing (RIP/OSPF)
Pros	Pros
1. Easy to configure for small networks.	1. Automatically updates routes when the network changes.
2. Uses less CPU and bandwidth because no routing updates are exchanged.	2. Best for large networks with many routers.
3. More secure because routes are entered manually.	3. Quickly finds an alternate path if one route fails.
Cons	Cons
1. Must be updated manually whenever the network changes.	1. More complex to configure and manage.
2. Not suitable for large networks.	2. Uses more CPU, memory, and bandwidth for routing updates.
3. If a link fails, traffic stops until the route is manually changed.	3. May take some time to converge after a network change.

Real-World Examples

- **Static Routing:** A small restaurant connected to one router can use static routing because the network rarely changes.
- **Dynamic Routing (OSPF/RIP):** A food delivery app like Zomato has many servers and branches. If one network path fails, dynamic routing automatically selects another route, ensuring the app continues to work without interruption.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

Router 1 (R1)

```
enable  
configure terminal  
router ospf 1  
network 192.168.1.0 0.0.0.255 area 0  
network 10.0.0.0 0.0.0.3 area 0  
end  
write memory
```

Router 2 (R2)

```
enable  
configure terminal  
router ospf 1  
network 192.168.2.0 0.0.0.255 area 0  
network 10.0.0.0 0.0.0.3 area 0  
end  
write memory
```

Command	Explanation
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
router ospf 1	Starts the OSPF routing process with process ID 1.
network 192.168.1.0 0.0.0.255 area 0	Enables OSPF for the 192.168.1.0/24 network in Area 0 (Router 1).
network 192.168.2.0 0.0.0.255 area 0	Enables OSPF for the 192.168.2.0/24 network in Area 0 (Router 2).
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the link connecting the two routers.
end	Exits configuration mode.
write memory	Saves the configuration so it remains after a reboot.